

Swasthya Sanjivani: A Portable Digital Testing Laboratory

Swasthya Sanjivani is an automated and, most importantly, affordable solution to blood and urine testing with immediate confirmatory results. The system is basically a portable digital testing laboratory that aims at addressing the lack of diagnostic facilities in rural areas. It also consists of an electronic pill box, which reminds the patients about their medication and appointment schedule. Find out more about it

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India is the second-most-populated country in the world. Its rural and urban distribution of population is 68.84 per cent and 31.16 per cent, respectively, as per the Census Report 2011. Well, nothing much can be done about the growing population apart from educating people. But education, poverty and health in the rural areas need more focus. Swasthya Sanjivani—an innovative kit developed at Manav Rachna College of Engineering (MRCE)—aims to eliminate lack of healthcare facilities.

Trigger for innovation

There is a team of five behind Swasthya Sanjivani—Manasvi Sihag, Hardik Garg, Nikhil Dalmia, Prashant Gupta and Pritam Singh.

Before delving into what Swasthya Sanjivani is all about, let us look at what inspired them to come up with this solution and what exactly they are trying to solve.

Moved by the fact that “One billion people lack access to basic healthcare,” the team decided to come up with a solution. They conducted a survey in 23 different villages, which revealed that these areas lacked adequate healthcare facilities, and people could not afford the current facilities because of lack of money.

The team says, “Even if these people get tested at small camps, they have to travel to distant locations to get further tests carried out, which puts a heavy burden on their restricted earnings.

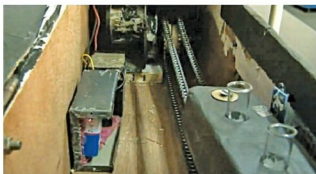


Fig. 1: Inside the kit (Prototype)

What is Swasthya Sanjivani?

Swasthya Sanjivani enables basic digital literates at isolated centres to perform tests prescribed by medical practitioners of repute at distant locations. These expert doctors, through video conferencing, would be present all the time, right from the time the patient enters the centre till he leaves it with all the answers concerning his health.

The system is basically a portable digital testing laboratory. It also consists of an electronic pill box, which is provided to the patients to remind them about their medicine and appointment schedule.

This project aims at addressing the lack of diagnostic facilities at a primary level, especially in rural areas. It could be termed as a cheaper, effective substitute for telemedicine. In telemedicine, the diagnosis can be performed from a remote location but it is an expensive process and demands an accurate setup. Swasthya Sanjivani is a simplified substitute as it requires

just the kit, Internet connection, trained technicians, and a laptop or PC.

The direct beneficiaries of this system are patients of anaemia, diabetes, renal damage and jaundice, people living in unapproachable areas, people below poverty line,

diagnostic centres and hospitals in heavily populated areas, and clinics into blood and urine testing.

System architecture

Swasthya Sanjivani has a three-tier architecture comprising:

Tier 1: Laptop. A laptop acts as the interface between the doctor and the Swasthya Sanjivani kit. Video conferencing connecting the doctor to the patient is achieved through Skype connection on the laptop.

Tier 2: Swasthya Sanjivani kit. This kit is supplied to operators in rural areas. The kit ensures that patients get an economical and on-the-spot blood and urine testing system at their doorstep.

Tier 3: Web portal (SkyDrive). The Web portal bridges the gap between rural and urban areas. The patient can be treated by a specialist doctor from a distant location.

How does it work?

The Swasthya Sanjivani kit works on the principle of colour-matching technique that has been widely used for urine and